

Phase 5 - Backups & Package/Patch Management

This document covers the backup and archiving workflow for critical configuration data, and treats patch management as an active security control rather than routine maintenance.

1. Backup Archive

A full backup of /etc was created using tar with xz compression, combining archiving, permission preservation, and compression into a single command. The output filename was timestamped automatically via command substitution:

```
total 3.7M
-rw-r--r--. 1 root root 3.7M Jul  4 20:45 etc-backup-2026-07-04.tar.xz
```

1.1 Compression Result

The original /etc directory measures approximately 30MB uncompressed, compared to the final 3.7MB xz-compressed archive — roughly an 8:1 reduction, consistent with xz's reputation as the strongest (though slowest) of the standard Linux compression tools:

```
30M/etc
```

1.2 Archive Contents Verification

The archive was verified without extraction using tar -tJf, confirming the expected system configuration files and directories are present, including fstab, crypttab, and the full LVM configuration tree:

```
etc/
etc/fstab
etc/crypttab
etc/lvm/
etc/lvm/devices/
etc/lvm/devices/system.devices
etc/lvm/archive/
etc/lvm/archive/rhel_00000-1156436274.vg
etc/lvm/backup/
etc/lvm/backup/rhel
etc/lvm/cache/
etc/lvm/profile/
etc/lvm/profile/cache-mq.profile
etc/lvm/profile/thin-generic.profile
etc/lvm/profile/vdo-small.profile
```

2. Patch Management as a Security Control

Rather than treating updates as routine housekeeping, dnf update --security was used to scope patching specifically to security-relevant packages:

```
Updating Subscription Management repositories.
Last metadata expiration check: 0:04:10 ago on Sat 04 Jul 2026 08:54:45 PM EEST.
Dependencies resolved.
Nothing to do.
Complete!
```

"Nothing to do" confirms the system had no outstanding security patches pending at the time of the check — an expected and healthy result on a recently provisioned system.

2.1 dnf Transaction History (Audit Trail)

dnf's built-in transaction history was reviewed as a lightweight audit trail of every package operation performed on the system:

ID	Command line	Date and time
Action(s)	Altered	

3	update --security	2026-07-04 20:48
I, U	97	
2	install -y vim tar bzip2 xz chrony firewalld cockpit	2026-07-03 18:45
Upgrade	8	
1		2026-06-24 13:47
Install	1343	

Note: Every transaction is numbered and reversible via `dnf history undo <ID>`, a meaningful safety net over raw rpm operations, which have no equivalent built-in rollback mechanism.

3. Package Inventory

A quick inventory count of all installed packages was taken as a baseline reference point:

```
1328
```

4. Reverse Lookup: Package Owning sshd

To confirm exactly which package provides the hardened SSH daemon binary (relevant given the Phase 3 hardening work), a reverse lookup was performed against `/usr/sbin/sshd`. Multiple historical package versions are returned by `dnf provides` since it searches all available repository metadata, not just what's installed — the version actually installed on this system is marked `@System` in the `repo` field:

```
openssh-server-9.9p1-23.el10_2.x86_64 : An open source SSH server daemon
Repo      : @System
Matched from:
Filename   : /usr/sbin/sshd
```

This confirms the currently installed SSH daemon is `openssh-server` version `9.9p1-23.el10_2`, a small but genuine software-inventory check relevant to vulnerability management — knowing precisely which package and version owns a critical binary is a prerequisite for assessing exposure to any newly disclosed CVE.